

Context: You're working for a rather quirky boss who's set on being the next big AI company. AI company that does what you may ask? Those are unimportant details, entrepreneurs don't think about that stuff, they just iterate. So your boss, "gifted" in the arts of vibe coding, has made 3 different prototypes for his future unicorn business. A video game, an ecommerce site, and a data analytics dashboard. Like we said, well scoped business propositions are for chumps. Your boss, fresh off his latest trip to Bali/Bora Bora/Belize, informs you he "vibe coded 3 possibilities for the next big thing" on his flight back. He wants you to take each of the options and add "sparkle" before your investor meeting at the end of the week. You open each of the three options and.. well... They do run, so there's that...

Description:

We've uploaded three zip files of your bosses projects to canvas.

For each of the 3 projects, you need to:

1. **Fix the Apps.** There are currently bugs in all three of the apps. Use your AI agent to hunt down and fix the bugs in the apps so they are usable and pass the provided ``public_tests/`.*`
2. **Build Additional Features:** Implement the missing business features for all 3 apps.
 - a. *Game:*
 - i. Users select a character image to play with.
 - ii. A "power up", where the player eats a mushroom and gets double points for 10 seconds.
 - b. *Dashboard:*
 - i. Optional filter by data-range that is applied to all the charts.
 - ii. Select 5 of 8 chart options to be displayed after the csv file is uploaded.
 - c. *eCommerce:*
 - i. If the user has ≥ 2 pairs of the *light wash* jeans in the cart, a belt is automatically added as a free item.
 - ii. Estimate shipping time using the customers zip code with values that you (the developer) provide
3. **Secure the Edge Cases:** You are only seeing a fraction of the test cases. You must prompt your AI to anticipate edge cases (e.g., negative numbers, malicious strings) and handle them gracefully.

Deliverable:

A single ``zip`` file of each of the codebases (or a GitHub repo link). You **MUST** export the code; do not submit just a URL. No essays or write-ups are required. (*Hint: ask Claude how to zip your folder from the terminal.*)

Grading

Grades will be based on:

- The tests we have given you passing (without changing them)
- The tests we've held back passing
 - This will primarily cover the edge cases we've asked you to think about.
- The additional features are functional and well thought out

* This is a technical term, don't worry if you don't know it! You can always ask an LLM to explain it to you!

Getting Started

You'll be running these projects at least partially in your terminal. You can use Claude Code in your terminal or in the desktop app.

*Hint: If you have any questions we **really** encourage you to try asking the LLM a few times before you reach out to the TAs - the process of learning how to leverage LLMs as a personal tutor will pay off!*

Install Dependencies:

Node/npm: <https://docs.npmjs.com/downloading-and-installing-node-js-and-npm>

Claude Code in Terminal: <https://code.claude.com/docs/en/terminal-guide>

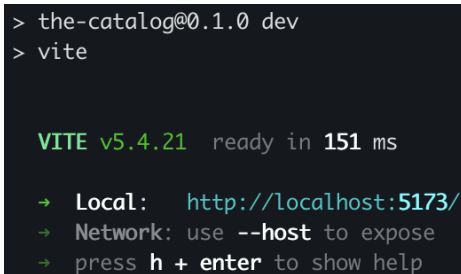
How to Run the App

In your terminal ([mac](#), [windows](#)), you'll be using the following commands (aka instructions) to run your app and run the tests. Copy and paste the instructions directly into your terminal and hit enter.

Getting dependencies needed to run the App (YOU NEED TO RUN THIS FIRST): npm install

Starting the App: npm run dev

- This will give you a local host url to go to to see your app on your computer!
- It will look something like this after you run

A terminal window with a dark background. The prompt is 'the-catalog@0.1.0 dev'. The user has entered 'vite'. The output shows 'VITE v5.4.21 ready in 151 ms'. Below this, there are three lines of information: 'Local: http://localhost:5173/', 'Network: use --host to expose', and 'press h + enter to show help'.

```
> the-catalog@0.1.0 dev
> vite

VITE v5.4.21 ready in 151 ms

➔ Local:   http://localhost:5173/
➔ Network: use --host to expose
➔ press h + enter to show help
```

- Note: the port (ie. 5173 in ^^, might be different for you, but that doesn't matter)
- Press Ctrl C to stop the app from running

Running Tests: npm run test

Tip: It might be nice to have 3 terminal windows open for this to be able to run all the processes without having to exit.

A note on how “Technical” this assignment is

This assignment definitely gets you closer to writing code/technical implementation than past assignments, but Claude is very good at acting as the “Developer” to your Product Manager. Claude will likely show you a lot of code as it goes or use technical terms you don't know, but DON'T WORRY! This is a great opportunity to ask Claude to explain things in non-technical terms or tell it you're non-technical when you start vibe coding so it can adapt accordingly.